Classification Lecture 9

Friday, 11 October 2024 6:08 AM

Regularization in Logistic Regression

↳ helps in reducing model overfitting. It adds a penalty term which encourages simpler models, leading to generalization on unseen data.

Lasso Regularization (L1 norm)

$$J(\beta) = \underbrace{-\log L(X;\beta)}_{-\log \text{ likeihood (cost function)}} + \lambda \|\beta\|_1$$

max log likelihood

min − log likelihood

$$L(X;\beta) = \prod_{i=1}^{N} p(x_i)^{y_i} \cdot (1-p(x_i))^{1-y_i}$$

Updated Cost function:-

$$J(\beta) = -\left[\sum_{i=1}^{N} y_i \log p(x_i) + (1-y_i)\log(1-p(x_i))\right] + \lambda \|\beta\|_1$$

$$= -\left[\sum_{i=1}^{N}\left[-\log(1+e^{x_i\beta})\right] + \sum_{i=1}^{N} y_i x_i \beta\right] + \underbrace{\lambda \sum_{i=1}^{N} |\beta_i|}_{L_1 \text{ penalty}}$$

$$\frac{\partial}{\partial \beta_i} |\beta_i| = \text{sign}(\beta_i)$$

$$\frac{\partial J(\beta)}{\partial \beta_0} = -\sum_{i=1}^{N} (y_i - \sigma(z_i)) + \lambda \cdot \text{sign}(\beta_0)$$

$$\frac{\partial J(\beta)}{\partial \beta_k} = -\sum_{i=1}^{N} (y_i - \sigma(z_i)) \cdot x_{ki} + \lambda \cdot \text{sign}(\beta_k)$$

Since this is a cost function, we can apply gradient descent to calculate β values.

<u>Ridge Regularization</u> (L_2 norm)

$$J(\beta) = \underbrace{-\log L(X;\beta)}_{-\log \text{ likeihood (cost function)}} + \underbrace{\lambda \|\beta\|_2^2}_{L_2 \text{ penalty}}$$

Updated Cost function:-

$$J(\beta) = -\left[\sum_{i=1}^{N} y_i \log p(x_i) + (1-y_i) \log (1-p(x_i))\right] + \lambda \|\beta\|_2^2$$

$$= -\left[\sum_{i=1}^{N} [-\log(1+e^{x_i \beta})] + \sum_{i=1}^{N} y_i x_i \beta\right] + \underbrace{\lambda \sum_{i=1}^{N} \beta_i^2}_{L_1 \text{ penalty}}$$

$$\frac{\partial}{\partial \beta_i} \beta_i^2 = 2\beta_i$$

$$\frac{\partial J(\beta)}{\partial \beta_0} = -\sum_{i=1}^{N} (y_i - \sigma(z_i)) + 2\lambda\beta_0$$

$$\frac{\partial J(\beta)}{\partial \beta_k} = -\sum_{i=1}^{N} (y_i - \sigma(z_i)) \cdot x_{ki} + 2\lambda\beta_k$$

Multi-class classification using Logistic Regression

$$P(y=k \mid \vec{x}) = \text{softmax}(z_k)$$

$$= \frac{e^{xw_k}}{\sum_{i=1}^{k} e^{xw_i}} = \frac{e^{w_{0k} + w_{1k} \cdot x_1 + \dots + w_{pk} \cdot x_p}}{\sum_{i=1}^{k} e^{w_{0i} + w_{1i} \cdot x_1 + \dots + w_{pi} \cdot x_p}}$$

$$\# \text{ weights} = k \times (p+1)$$

$$x \cdot w_k$$

x: $1 \times (p+1)$ $\quad$ w_k: $(p+1) \times 1$

Log-likelihood :-

$$l(X, W) = \sum_{i=1}^{N} \sum_{j=1}^{K} \underbrace{1 \cdot \{y_i = k\}}_{} \log P\{y_i = k \mid x_i\}$$

indicator function, which is 1 if $y_i = k$, and 0 otherwise

0 otherwise

	$\vec{x}$				y (multi-class)
	x_1	x_2	$\cdots$	x_p	$1 \cdots K$
1	x_{11}	x_{21}		x_{p1}	1
2	$\vdots$				7
$i^{th} \rightarrow$ 3	$\cdot$	$\cdot \; \cdot \; \cdot$		$\cdot$	k
$\vdots$					K
$\vdots$				$\cdot$	$\vdots$
N	x_{1N}	x_{2N}	$\cdots$	x_{pN}	

		x_1	x_2	y (1/2/3)
$x_1 \rightarrow$	1	3	1	1 $\leftarrow$
x_2	2	4	2	2 $\leftarrow$
x_3	3	7	5	3 $\leftarrow$

$$l = \log P(y=1 \mid x_1) + \log P(y=2 \mid x_2) + \log P(y=3 \mid x_3)$$

$$P(y=1 \mid x_1) = \frac{e^{x_1 \cdot w_1}}{e^{x_1 \cdot w_1} + e^{x_1 \cdot w_2} + e^{x_1 \cdot w_3}}$$

$$P(y=2 \mid x_2) = \frac{e^{x_2 \cdot w_2}}{e^{x_2 \cdot w_1} + e^{x_2 \cdot w_2} + e^{x_3 \cdot w_3}} \cdots$$

> log-likelihood is a complex function of w_i vectors, we use gradient-ascent to compute w_i for every class.

eg. We use logistic regression to compute the class of an unknown data point in 3-class classification problem. the logits for the classes A, B, C are $z_A = 3$, $z_B = 2.5$, $z_C = 4$. Find the probabilities of class A, B and C.

Sol.

$$P(A) = \text{softmax}(z_A) = \frac{e^{z_A}}{e^{z_A} + e^{z_B} + e^{z_C}} = \frac{e^3}{e^3 + e^{2.5} + e^4} \approx 0.228$$

$$P(B) = \frac{e^{2.5}}{e^3 + e^{2.5} + e^4} \approx 0.138, \quad P(C) \approx 0.638$$

=x=

eg.

Age	Nominal Gender	salary	y (A/B/C)
10	M|F|N	10K	A
20	[1, 0, 0]	1 Lakh	A
25	[0, 1, 0]	1.5Cr	B
⋮	[0, 0, 1]	⋮	C
⋮			C
			⋮

		isMale	isFemal	isNotAppl.		y
1 →	10	1	0	0	10 k	A
2 →	20	0	1	0	1 lakh	A
3 →	25	0	0	1	1.5Cr	B
⋮	⋮	⋮	⋮	⋮	⋮	⋮
1000						

$$\frac{e^{w_{0A} + 10\times w_{1A} + 1\times w_{2A} + 0\times w_{3A} + 0\times w_{4A} + 10K \times w_{5A}}}{\Sigma e \cdots}$$

$$\left.\begin{array}{l} W_A : \; w_{0A}, \; w_{1A} \; \cdots, \; w_{5A} \\ w_B : \; w_{0B}, \; w_{1B} \; \cdots \; w_{5B} \\ w_C : \; w_{0C}, \; w_{1C} \; \cdots \; w_{5C} \end{array}\right\} \text{18 weights}$$

35, 0, 1, 0, 25k

$$\begin{cases} Z_A = w_{0A} + 35\times w_{1A} + 0\times w_{2A} + 1\times w_{3A} + 0\times w_{4A} + 25k\times w_{5A} \\ Z_B = \cdots \qquad Z_C = \cdots \end{cases}$$